

Grade 8
Unit 10
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

For problems 1–4, determine if each table could represent a linear function.
Explain your reasoning.

Relation	Description										
1. <table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>-4</td><td>-7</td></tr> <tr> <td>-3</td><td>-5</td></tr> <tr> <td>2</td><td>5</td></tr> <tr> <td>3</td><td>7</td></tr> </tbody> </table> $\frac{\Delta y}{\Delta x} = 2$	x	y	-4	-7	-3	-5	2	5	3	7	The relation ☐ could be a linear function because ... ☐ cannot be the rate of change is constant between each set of coordinates.
x	y										
-4	-7										
-3	-5										
2	5										
3	7										
2. <table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>-3</td><td>14</td></tr> <tr> <td>-1</td><td>4</td></tr> <tr> <td>4</td><td>-21</td></tr> <tr> <td>5</td><td>-24</td></tr> </tbody> </table> $\frac{\Delta y}{\Delta x} = \frac{-21 - (-24)}{4 - 5} = -3$	x	y	-3	14	-1	4	4	-21	5	-24	The relation ☐ could be a linear function because ... ☐ cannot be the rate of change is not constant between each set of coordinates.
x	y										
-3	14										
-1	4										
4	-21										
5	-24										
3. <table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>-15</td><td>-11</td></tr> <tr> <td>-5</td><td>-16</td></tr> <tr> <td>0</td><td>-21</td></tr> <tr> <td>5</td><td>-26</td></tr> </tbody> </table> $\frac{\Delta y}{\Delta x} = \frac{-16 - (-21)}{-5 - 0} = -1$	x	y	-15	-11	-5	-16	0	-21	5	-26	The relation ☐ could be a linear function because ... ☐ cannot be the rate of change is not constant between each set of coordinates.
x	y										
-15	-11										
-5	-16										
0	-21										
5	-26										
4. <table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr> <td>-5</td><td>3</td></tr> <tr> <td>-3</td><td>18</td></tr> <tr> <td>-1</td><td>108</td></tr> <tr> <td>1</td><td>648</td></tr> </tbody> </table> $\frac{\Delta y}{\Delta x} = \frac{108 - 18}{-1 - (-3)} = 45$	x	y	-5	3	-3	18	-1	108	1	648	The relation ☐ could be a linear function because ... ☐ cannot be the rate of change is not constant between each set of coordinates.
x	y										
-5	3										
-3	18										
-1	108										
1	648										

$$\frac{\Delta y}{\Delta x} = \frac{14 - 4}{-3 - (-1)} = -5$$

$$\frac{\Delta y}{\Delta x} = \frac{-11 - (-16)}{-15 - (-5)} = -0.5$$

$$\frac{\Delta y}{\Delta x} = \frac{3 - 18}{-5 - (-3)} = 7.5$$

For problems 5–10, determine if each relation represents a linear function.

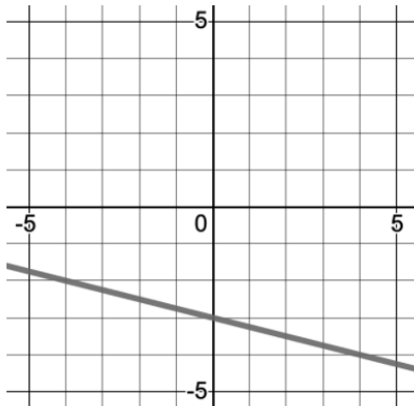
5. $y = 4 - 2x$

Linear, the equation is written in the form $y = mx + b$.

6. $y = 3^x + 4$

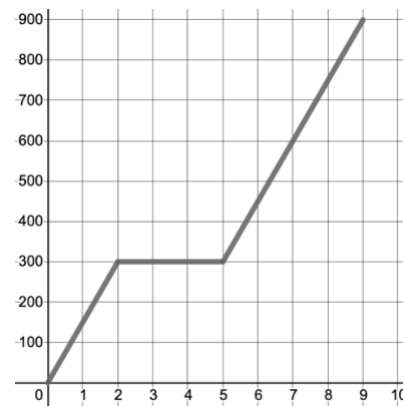
Nonlinear, the equation cannot be written in the form $y = mx + b$.

7.



Linear
The equation makes a line when graphed.

8.



Nonlinear
The equation does not make a line when graphed.

9. $\frac{x}{y} = 1$

Linear, because the equation can be written in the form $y = mx + b$.

10. $\frac{y}{5} - \frac{x}{4} = \frac{1}{3}$

Linear, because the equation can be written in the form $y = mx + b$.